R Notebook

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(stringr)
library(ggplot2)
library(RColorBrewer)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.0.4

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(knitr)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(rstatix)
```

```
##
## Attaching package: 'rstatix'
##
## The following object is masked from 'package:stats':
##
##     filter
```

```r
library(coin)
```

```
## Loading required package: survival
##
## Attaching package: 'coin'
##
## The following objects are masked from 'package:rstatix':
##
##     chisq_test, friedman_test, kruskal_test, sign_test, wilcox_test
```

```r
library(patchwork)
library(ggpubr)
library(viridis)
```

```
## Loading required package: viridisLite
```

```r
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
knitr::opts_chunk$set(dev = "cairo_pdf")
```

```r
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_al
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_me

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# Pond sample names
pond_samples = c("s9","s10","s19","s20")

# Create sample metadata using helper function
sample_metadata = create_sample_metadata(seq_metrics, pond_samples) %>%
  select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Extract HPV data before filtering
hpv_data = merged_data %>%
  filter(experiment == 1 | experiment == 2) %>%
  filter(str_detect(name, regex("Alphapapillomavirus", ignore_case = TRUE)))
pcv_data = merged_data %>%
  filter((experiment == 1 & library_prep == "MSSPE")) %>%
  filter(str_detect(name, regex("Circovirus sp.", ignore_case = TRUE)))
```

```r
# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(nt_rpm >= 1) %>% # so zeros don't skew comparison
  #filter((experiment == 1 & library_prep == "NGS") | (experiment == 2 | experiment == 3)) %>% # exp 1
  filter(water_control == "No") %>%   # filter out Negative Controls
  filter(infection_class != "Definite") %>%   # filter out positive control samples
  add_row(hpv_data)# %>%   # but keep hpv data
  #add_row(pcv_data) # and pcv msspe data from exp 1
```

```r
# Assign standardized names for viral targets
name_targets = c(
  "PBoV",   # Porcine bocavirus
  "PCV",    # Porcine circovirus 2
  "PKV",    # Porcine kobuvirus / Aichivirus C
  "EV-G",   # Porcine enterovirus G
  "PTeV",   # Porcine teschovirus A
  "PAstV",  # Porcine astrovirus 4
  "PSapoV", # Porcine sapovirus
  "PSapeV", # Porcine sapelovirus
  "HPV18"   # Human papillomavirus 7/18
)

# Filter to viral category and exclude phages
all_viruses <- all_data_species %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false") %>%
  filter(!str_detect(name, paste(c("Siphoviridae","Podoviridae","Myoviridae"), collapse = "|")))

# Add standardized name groups using helper function
all_viruses = add_viral_name_groups(all_viruses)

# Small constant for log transformation of zero RPM values
eps <- 0.1

# Filter to only the target viral species or genus
only_viral_targets = all_viruses %>%
  group_by(base_sample, library_prep, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore
  summarize(rpm_sum = sum(nt_rpm, na.rm=TRUE), .groups="drop") %>% # sum rPM for each target
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_sum
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0.1), # Set NAs to 0.1 for non-detections
    NGS   = replace_na(NGS, 0.1),   # Set NAs to 0.1 for non-detections
    rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))
```

```r
# calculate average enrichment for all targets
mean_enrich_overall = only_viral_targets %>%
  summarise(
    mean_log2FC = mean(log2FC),
    median_foldchange = median(rpm_enrichment),
    mean_foldchange = mean(rpm_enrichment))

mean_enrich_overall
```

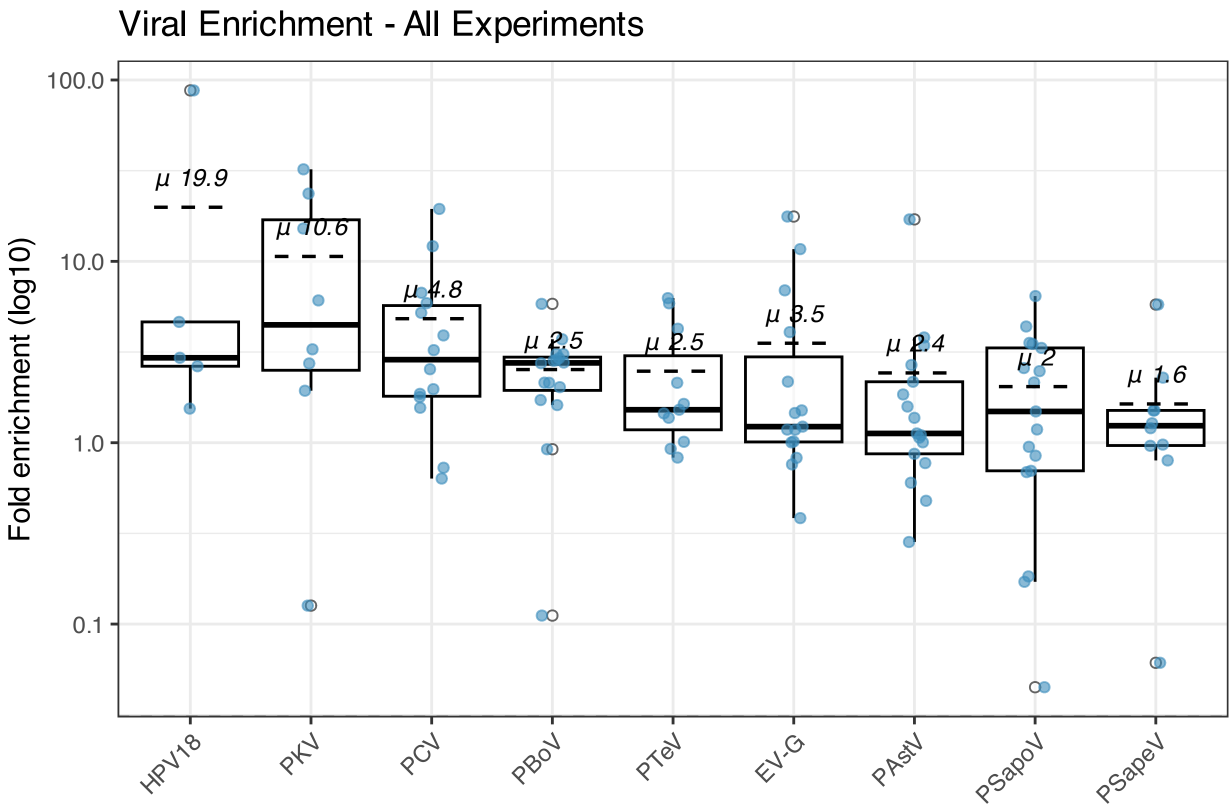
```
## # A tibble: 1 x 3
##   mean_log2FC median_foldchange mean_foldchange
##         <dbl>             <dbl>           <dbl>
## 1       0.940              1.86            4.12
```

```r
# calculate mean so we can add to the boxplot
mean_df <- only_viral_targets %>%
  group_by(name_group) %>%
  summarise(mean_value = mean(rpm_enrichment, na.rm = TRUE), median_value = median(rpm_enrichment, na.r

# plot fold change across all samples
alltarget_boxplot <- ggplot(only_viral_targets, aes(x = name_group, y = rpm_enrichment)) +
  geom_boxplot(width = 0.8, outlier.shape = 21, outlier.fill = "white", alpha = 0.6, width = 0.6, color
  scale_y_log10() +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, color = "#3C8DBC") +  # Professional blue
  geom_hline(yintercept = 0, linetype = "dashed") +
  #dotted mean line
  geom_segment(
    data = mean_df,
    aes(x = as.numeric(name_group) - 0.3,
        xend = as.numeric(name_group) + 0.3,
        y = mean_value, yend = mean_value),
    inherit.aes = FALSE,
    linetype = "dashed", color = "black", linewidth = 0.6
  ) +
  geom_text(data = mean_df, aes(x = name_group, y = mean_value, label = paste0("  ", round(mean_value, 1
  scale_x_discrete() +  # adds space between boxes for labels
  labs(
    title = "Viral Enrichment - All Experiments",
    x = "",
    y = "Fold enrichment (log10)",
  ) +
  theme_bw() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1),
  )
```

```
## Warning: Duplicated aesthetics after name standardisation: width
```

```r
alltarget_boxplot
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```




```r
# Do calculations for reads/nt_count instead of rpm
reads_summary = all_viruses %>%
  group_by(base_sample, library_prep, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(reads_sum = sum(nt_count, na.rm=TRUE), .groups="drop") %>% # sum rPM for each target
  pivot_wider(
    names_from = library_prep,
    values_from = reads_sum
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 1), # Set NAs to 0 for non-detections
    NGS   = replace_na(NGS, 1),   # Set NAs to 0 for non-detections
    reads_enrichment = (`MSSPE` + 1)/(`NGS` + 1),
    log2FC = log2(reads_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))
reads_summary
```

```
## # A tibble: 113 x 6
##    base_sample name_group MSSPE   NGS reads_enrichment log2FC
##    <chr>       <fct>      <int> <int>            <dbl>  <dbl>
##  1 s7          PBoV        3023    46             64.3   6.01
##  2 s7          PCV          690    10             62.8   5.97
##  3 pc5         HPV18        108     1             54.5   5.77
##  4 s4          PCV         1565    35             43.5   5.44
##  5 s4          PBoV         350     9             35.1   5.13
##  6 s8          EV-G        2062    69             29.5   4.88
```

```
##   7 s7          PKV          262     8             29.2   4.87
##   8 s8          PCV          429    14             28.7   4.84
##   9 s2          PCV          241     8             26.9   4.75
##  10 s8          PTeV         250    10             22.8   4.51
## # i 103 more rows
```

```
overall_reads_enrichment = reads_summary %>%
  group_by(name_group) %>%
  summarize(
    median_FC = median(reads_enrichment),
    mean_FC   = mean(reads_enrichment),
    median_log2FC = median(log2FC),
    n_pairs = n(),
    .groups = "drop"
  ) %>%
  arrange(desc(median_log2FC))
overall_reads_enrichment
```

```
## # A tibble: 9 x 5
##   name_group median_FC mean_FC median_log2FC n_pairs
##   <fct>          <dbl>   <dbl>         <dbl>   <int>
## 1 PKV            12.7    11.8          3.66        8
## 2 PCV             3.86   14.3          1.95       14
## 3 PSapoV          2.94    4.80         1.55       17
## 4 PBoV            2.88    9.11         1.52       16
## 5 HPV18           2.18   14.5          1.12        5
## 6 PTeV            2       5.72         1          11
## 7 EV-G            1.99    5.83         0.991      15
## 8 PAstV           1.82    4.52         0.860      17
## 9 PSapeV          1.45    2.79         0.530      10
```

Viral Enrichment w/o primers from experiment 1

```
# Create separate dataset for experiment 1 analysis
merged_data_exp1 <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species_exp1 <- merged_data_exp1 %>%
  filter(tax_level == 1) %>%
  filter(experiment == 1) %>%  # Only experiment 1
  filter(water_control == "No") %>%   # Exclude negative controls
  filter(!base_sample %in% c("pc1","pc3")) %>%  # Exclude zymo fecal controls
  filter(infection_class != "Definite")   # Exclude positive controls
  #filter(nt_rpm >= 1) # rpm at least 1

# Filter to viral category and exclude phages
all_viruses_exp1 <- all_data_species_exp1 %>%
  filter(category == "viruses") %>%
  filter(is_phage == "false")

# Add standardized name groups using helper function
all_viruses_exp1 = add_viral_name_groups(all_viruses_exp1)
```

```r
# Filter to only the target viral species or genus
only_viral_targets_exp1 = all_viruses_exp1 %>%
  group_by(base_sample, library_prep, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(rpm_sum = sum(nt_rpm, na.rm=TRUE), .groups="drop") %>%
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_sum
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0.1), # Set NAs to 0 for non-detections
    NGS   = replace_na(NGS, 0.1),   # Set NAs to 0 for non-detections
    rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))

# calculate average enrichment for all targets
mean_enrich_overall_exp1 = only_viral_targets_exp1 %>%
  filter(name_group != "PToV*") %>%
  summarise(mean_log2FC = mean(log2FC), median_foldchange = median(rpm_enrichment))

mean_enrich_overall_exp1
```

```
## # A tibble: 1 x 2
##   mean_log2FC median_foldchange
##         <dbl>             <dbl>
## 1       0.255              1.15
```

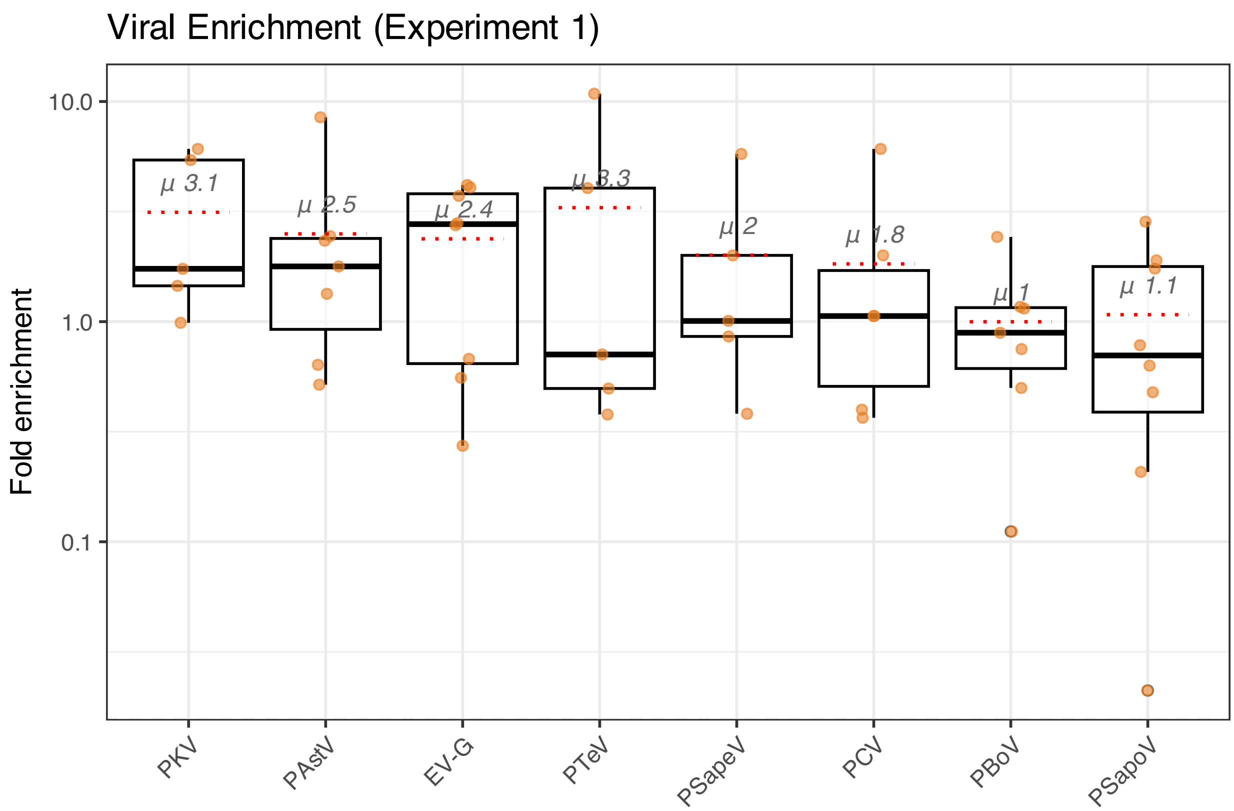
```r
# calculate mean so we can add to the boxplot
mean_df_exp1 <- only_viral_targets_exp1 %>%
  group_by(name_group) %>%
  summarise(mean_value = mean(rpm_enrichment, na.rm = TRUE))

# plot fold change across all samples
alltarget_boxplot_exp1 <- ggplot(only_viral_targets_exp1, aes(x = name_group, y = rpm_enrichment)) +
  geom_boxplot(width = 0.8, outlier.shape = 21, outlier.fill = "white", alpha = 0.6, width = 0.6, color
  scale_y_log10() +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, color = "#E67E22") +  # Professional orange
  geom_hline(yintercept = 0, linetype = "dashed") +
  # dotted mean line
  geom_segment(
    data = mean_df_exp1,
    aes(x = as.numeric(name_group) - 0.3,
        xend = as.numeric(name_group) + 0.3,
        y = mean_value, yend = mean_value),
    inherit.aes = FALSE,
    linetype = "dotted", color = "red", linewidth = 0.6
  ) +
  geom_text(data = mean_df_exp1, aes(x = name_group, y = mean_value, label = paste0("  ", round(mean_val
  scale_x_discrete() +  # adds space between boxes for labels
  labs(
```

```
    title = "Viral Enrichment (Experiment 1)",
    x = "",
    y = "Fold enrichment",
  ) +
  theme_bw() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 45, hjust = 1),
  )
```

```
## Warning: Duplicated aesthetics after name standardisation: width
```

```
alltarget_boxplot_exp1
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

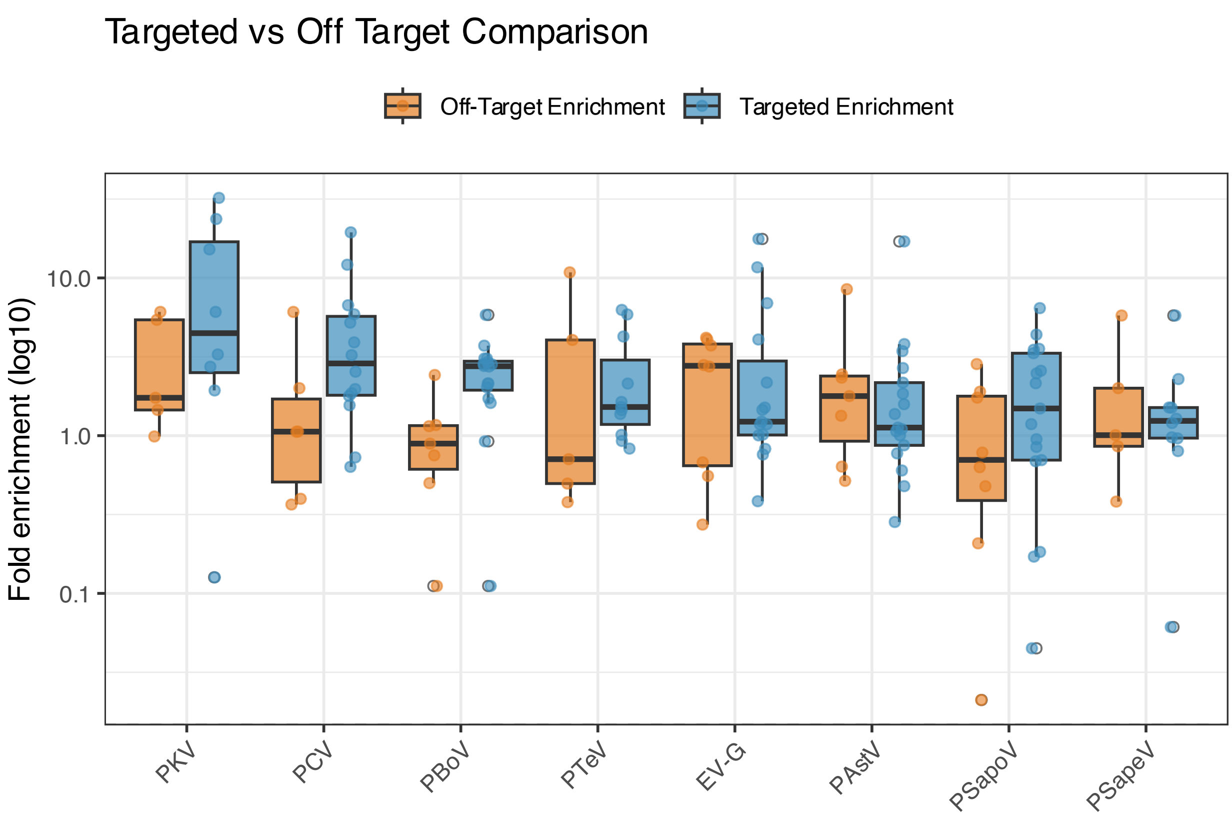


Combined Enrichment Plot

```
# Combine data from both experiments for unified visualization
combined_targets <- bind_rows(
  only_viral_targets %>% mutate(experiment_group = "Targeted Enrichment"),
  only_viral_targets_exp1 %>% mutate(experiment_group = "Off-Target Enrichment")
) %>%
  filter(!name_group == "HPV18")

# Create combined boxplot with faceting
```

```
combined_enrichment_plot <- ggplot(combined_targets, aes(x = name_group, y = rpm_enrichment, fill = expe
  geom_boxplot(width = 0.7, outlier.shape = 21, outlier.fill = "white", alpha = 0.7, position = positio
  scale_y_log10() +
  geom_point(aes(color = experiment_group),
             position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
             alpha = 0.6, size = 1.5) +
  scale_fill_manual(values = c("Targeted Enrichment" = "#3C8DBC", "Off-Target Enrichment" = "#E67E22"))
  scale_color_manual(values = c("Targeted Enrichment" = "#3C8DBC", "Off-Target Enrichment" = "#E67E22")
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(
    title = "Targeted vs Off Target Comparison",
    x = "",
    y = "Fold enrichment (log10)",
    fill = "Dataset",
    color = "Dataset"
  ) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "top",
    legend.title = element_blank()
  )

combined_enrichment_plot
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```



Final rPM Enrichment Tables for Manuscript

```r
cat("\\captionsetup{labelformat=empty}")
```

```r
# final table for each viral target
viral_target_enrichment_summary = only_viral_targets %>%
  group_by(name_group) %>%
  summarise(
    rpm_ngs = round(sum(NGS, na.rm = TRUE),2),
    rpm_msspe = round(sum(MSSPE, na.rm = TRUE),2),
    median_enrichment = round(median(rpm_enrichment, na.rm = TRUE),2),
    mean_enrichment = round(mean(rpm_enrichment, na.rm = TRUE),2),
    log_fold_enrichment = round(mean(log2FC, na.rm = TRUE),2),
    iqr = paste0(
      round(quantile(rpm_enrichment, 0.25, na.rm = TRUE), 1),
      "-",
      round(quantile(rpm_enrichment, 0.75, na.rm = TRUE), 1)
    ),
    n_pairs = n(),
    p_value = {
      # ensure both vectors are aligned and remove any NA pairs
      paired_data <- na.omit(data.frame(ngs = NGS, msspe = MSSPE))

      if(nrow(paired_data) > 1) {
        round(wilcox.test(paired_data$ngs,
                    paired_data$msspe,
                    alternative = "two.sided",
                    paired = TRUE)$p.value, 3)
      } else {
        NA_real_
      }
    },
  effect_r = {
      paired_data <- na.omit(data.frame(ngs = NGS, msspe = MSSPE))
      if(nrow(paired_data) > 1) {
        long_df <- paired_data %>%
          pivot_longer(cols = c(ngs, msspe), names_to = "prep", values_to = "value") %>%
          mutate(prep = factor(prep, levels = c("ngs","msspe")))
        round(wilcox_effsize(long_df, value ~ prep, paired = TRUE)$effsize, 3)
      } else NA_real_
    }
  ) %>%
  arrange(desc(median_enrichment))


# Make table pretty
viral_target_enrichment_summary %>%
  rename(
    "Virus" = name_group,
    "rPM (NGS)" = rpm_ngs,
    "rPM (MSSPE)" = rpm_msspe,
    "Median Fold Change" = median_enrichment,
    "Mean Fold Change" = mean_enrichment,
    "IQR" = iqr,
    "p-value" = p_value,
```

Enrichment across all paired samples per viral target

Virus	rPM (NGS)	rPM (MSSPE)	Median Fold Change	Mean Fold Change	IQR	p-value	effect size (r)
PKV	398.86	1,094.78	4.68	10.64	2.5–17.3	0.023	0.792
HPV18	24.48	82.53	2.94	19.88	2.6–4.6	0.062	0.905
PCV	298.03	779.39	2.89	4.83	1.8–5.7	0.005	0.713
PBoV	3,167.38	7,370.63	2.76	2.53	1.9–3	0.000	0.840
PTeV	378.22	520.99	1.52	2.48	1.2–3.2	0.032	0.643
PSapoV	603.32	1,343.98	1.49	2.04	0.7–3.3	0.132	0.373
PSapeV	191.88	272.98	1.24	1.64	1–1.5	0.322	0.338
EV-G	1,168.77	3,432.27	1.23	3.54	1–3.1	0.015	0.616
PAstV	3,118.68	6,067.98	1.12	2.43	0.9–2.2	0.329	0.247

```
    "effect size (r)" = effect_r
  ) %>%
  select(-n_pairs, -log_fold_enrichment) %>%
  kable(
    format = "latex",  # or "latex" if using LaTeX
    caption = "Enrichment across all paired samples per viral target",
    digits = 3,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )
```

Proportion of samples with viral targets

```
count_target_viruses = all_viruses %>%
  filter(!str_detect(sample_name, "PC")) %>%
  group_by(sample_name, name_group) %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  summarize(count = sum(n(), na.rm=TRUE), .groups="drop")

virus_sample_per_targets = count_target_viruses %>%
  group_by(name_group) %>%
  summarize(
    perc_of_samples = n_distinct(sample_name) / n_distinct(.$sample_name)
    ) %>%
  arrange(desc(perc_of_samples))
```

GENOME COVERAGE

```
cat("\\captionsetup{labelformat=empty}")
```

```
# Folder containing coverage CSVs
cov_dir <- "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_all/coverage_tables"

# Read all CSV files
```

```r
cov_files <- list.files(cov_dir, pattern = "\\.csv$", full.names = TRUE)

# Function to compute breadth and mean depth from one coverage file
calc_cov_stats <- function(file) {
  df <- read.csv(file)
  tibble(
    file = basename(file),
    virus = str_replace(file, "_s.*|_pc.*", ""),
    base_sample = str_remove(str_extract(basename(file), "(?<=_)(s\\d+|pc\\d+)"), "_"),
    prep = ifelse(str_detect(file, "_m"), "MSSPE", "NGS"),
    sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry"),
    ref_length = nrow(df),
    breadth_1x = mean(df$Coverage > 0) * 100,  # Percentage of genome covered at  1x depth
    mean_depth = mean(df$Coverage)  # Mean coverage depth across genome
  )
}

# Calculate coverage for all files
cov_summary <- map_dfr(cov_files, calc_cov_stats)

# Pivot wider for paired comparison
paired_tbl <- cov_summary %>%
  select(virus, base_sample, prep, breadth_1x, mean_depth) %>%
  pivot_wider(
    names_from = prep,
    values_from = c(breadth_1x, mean_depth),
    names_glue = "{.value}_{prep}"
  ) %>%
  mutate(
    enrichment_breadth = breadth_1x_MSSPE - breadth_1x_NGS,  # Absolute percentage point increase
    enrichment_depth   = mean_depth_MSSPE - mean_depth_NGS,  # Absolute depth increase
    virus = recode(virus,
      `astrov` = "PAstV",
      `etvg` = "EV-G",
      `hpv18` = "HPV18",
      `pbov` = "PBoV",
      `pcv` = "PCV",
      `sapel` = "PSapeV",
      `tev` = "PTeV",
      `pkv` = "PKV",
    )
  )

# Summary table (1 row per metric)
wilcox_summary = paired_tbl %>%
  replace_na(list(breadth_1x_MSSPE = 0, breadth_1x_NGS = 0, enrichment_breadth = 0)) %>% # Some samples
  group_by(virus) %>%
  summarise(
    mean_ngs = round(mean(breadth_1x_NGS, na.rm = TRUE),2),
    mean_msspe = round(mean(breadth_1x_MSSPE, na.rm = TRUE),2),
    percent_enrichment = round(mean_msspe - mean_ngs,2),
    iqr = paste0(
      round(quantile(breadth_1x_MSSPE - breadth_1x_NGS, 0.25, na.rm = TRUE), 1),
```

```
      "–",
      round(quantile(breadth_1x_MSSPE - breadth_1x_NGS, 0.75, na.rm = TRUE), 1)
    ),
    n_pairs = n(),
    p_value = if (n() > 1) {
      test <- wilcox.test(
        breadth_1x_MSSPE,
        breadth_1x_NGS,
        paired = TRUE,
        alternative = "two.sided",
        exact = FALSE
      )
      round(test$p.value, 3)
    } else {
      NA_real_
    },
    effect_r = {
      paired_data <- na.omit(data.frame(ngs = breadth_1x_NGS, msspe = breadth_1x_MSSPE))
      if(nrow(paired_data) > 1) {
        long_df <- paired_data %>%
          pivot_longer(cols = c(ngs, msspe), names_to = "prep", values_to = "value") %>%
          mutate(prep = factor(prep, levels = c("ngs","msspe")))
        round(wilcox_effsize(long_df, value ~ prep, paired = TRUE)$effsize, 3)
      } else NA_real_
    },
    .groups = "drop"
  )


# Make table pretty
wilcox_summary %>%
  mutate(
    virus = recode(virus,
      `astrov` = "PAstV",
      `etvg` = "EV-G",
      `hpv18` = "HPV18",
      `pbov` = "PBoV",
      `pcv` = "PCV",
      `sapel` = "PSapeV",
      `tev` = "PTeV",
      `pkv` = "PKV",
    )
  ) %>%
  rename(
    "Virus" = virus,
    "% coverage (NGS)" = mean_ngs,
    "% coverage (MSSPE)" = mean_msspe,
    "Enrichment (%)" = percent_enrichment,
    "IQR" = iqr,
    "p-value" = p_value,
    "Effect Size (r)" = effect_r
  ) %>%
  select(-n_pairs) %>%
```

Genome coverage enrichment across all paired samples per viral target

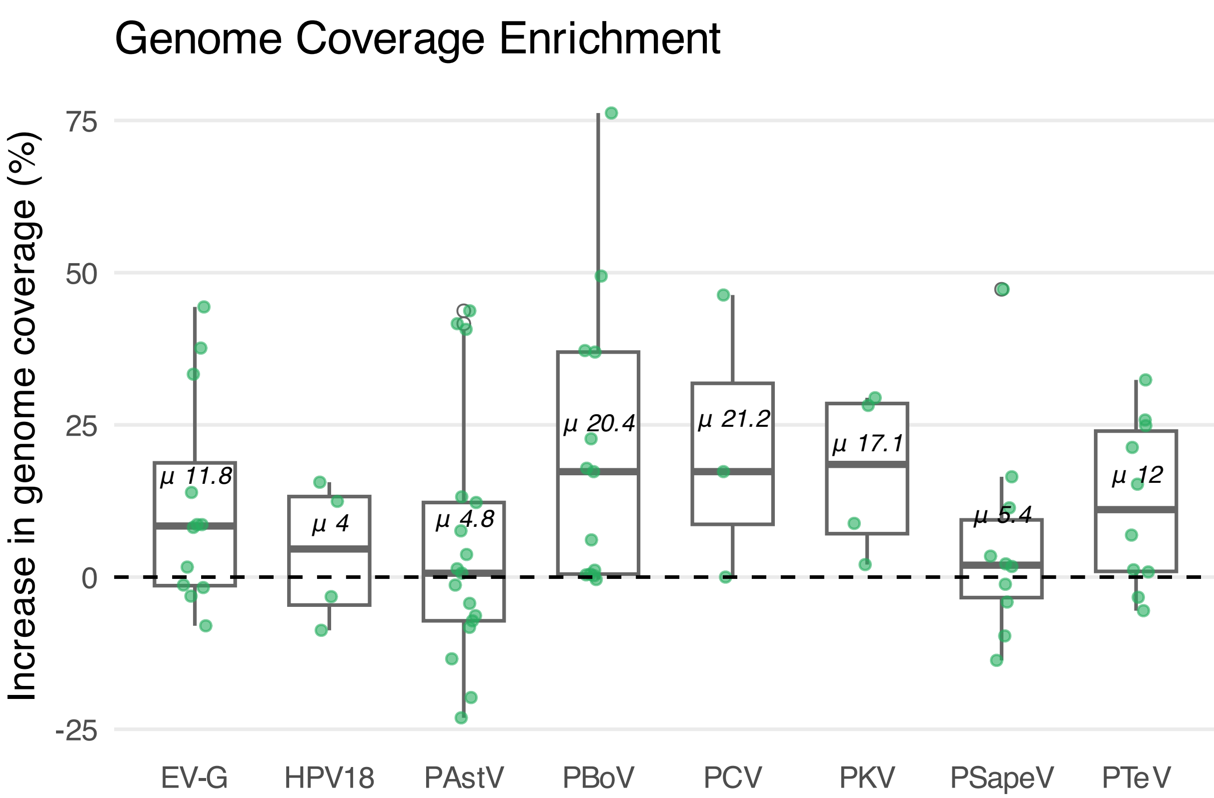
Virus	% coverage (NGS)	% coverage (MSSPE)	Enrichment (%)	IQR	p-value	Effect Size (r)
EV-G	40.96	48.62	7.66	-3.1–9.9	0.205	0.323
HPV18	15.23	19.25	4.02	-4.6–13.2	0.584	0.365
PAstV	52.43	57.14	4.71	-7–11.1	0.663	0.108
PBoV	56.90	74.51	17.61	0.4–29.8	0.006	0.719
PCV	34.17	56.66	22.49	17.6–28.2	0.059	0.861
PKV	11.44	20.60	9.16	2.3–9.4	0.019	0.757
PSapeV	32.81	36.37	3.56	-3.8–3.3	0.616	0.143
PTeV	33.49	43.40	9.91	1.2–21.3	0.025	0.630

```
  kable(
    format = "latex",  # or "latex" if using LaTeX
    caption = "Genome coverage enrichment across all paired samples per viral target",
    digits = 3,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )
```

```
# calculate mean so we can add to the boxplot
mean_cov_df <- paired_tbl %>%
  group_by(virus) %>%
  summarise(mean_value = mean(enrichment_breadth, na.rm = TRUE))

# Create the boxplot for each targeted virus
ggplot(paired_tbl, aes(x = virus, y = enrichment_breadth)) +
  geom_boxplot(width = 0.6, outlier.shape = 21, outlier.fill = "white", color = "gray40") +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, color = "#27AE60") +  # Professional green
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_segment(
    data = mean_cov_df,
    aes(x = as.numeric(virus) - 0.3,
        xend = as.numeric(virus) + 0.3,
        y = mean_value, yend = mean_value),
    inherit.aes = FALSE,
    linetype = "dashed", color = "#27AE60", linewidth = 0.6
  ) +
  geom_text(data = mean_cov_df, aes(x = virus, y = mean_value, label = paste0("  ", round(mean_value, 1)
  labs(
    x = "",
    y = "Increase in genome coverage (%)",
    title = "Genome Coverage Enrichment"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    panel.grid.major.x = element_blank(),
```

```
    panel.grid.minor = element_blank()
  )
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
## Warning in FUN(X[[i]], ...): NAs introduced by coercion

## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_segment()`).
```




```
# Overall Enrichment (non-paired)

# All viruses & enrichment
overall_tbl = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  group_by(name_group, library_prep) %>%
  summarize(rpm_mean = sum(nt_rpm, na.rm = TRUE), .groups = "drop")

overall_tbl_summary = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_
  group_by(base_sample, name_group, library_prep) %>%
```

```
  summarize(rpm_mean = sum(nt_rpm, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(
    names_from = library_prep,
    values_from = rpm_mean
  ) %>%
  mutate(
    MSSPE = replace_na(MSSPE, 0.1), # Set NAs to 0 for non-detections
    NGS   = replace_na(NGS, 0.1),   # Set NAs to 0 for non-detections
    rpm_enrichment = (`MSSPE` + eps)/(`NGS` + eps),
    log2FC = log2(rpm_enrichment),
    name_group = fct_reorder(name_group, log2FC, .fun = mean, .desc = TRUE)
  ) %>%
  arrange(desc(log2FC))

# Calculate average enrichment overall
median(overall_tbl_summary$log2FC)
```

```
## [1] 0.8947968
```

```
round(quantile(overall_tbl_summary$log2FC, 0.25, na.rm = TRUE), 2)
```

```
##  25%
## 0.02
```

```
round(quantile(overall_tbl_summary$log2FC, 0.75, na.rm = TRUE), 2)
```

```
##  75%
## 1.78
```

```
# median enrichment overall
median(overall_tbl_summary$rpm_enrichment)
```

```
## [1] 1.859348
```

```
# mean
mean(overall_tbl_summary$rpm_enrichment)
```

```
## [1] 4.116704
```

```
# IQR
round(quantile(overall_tbl_summary$rpm_enrichment, 0.25, na.rm = TRUE), 2)
```

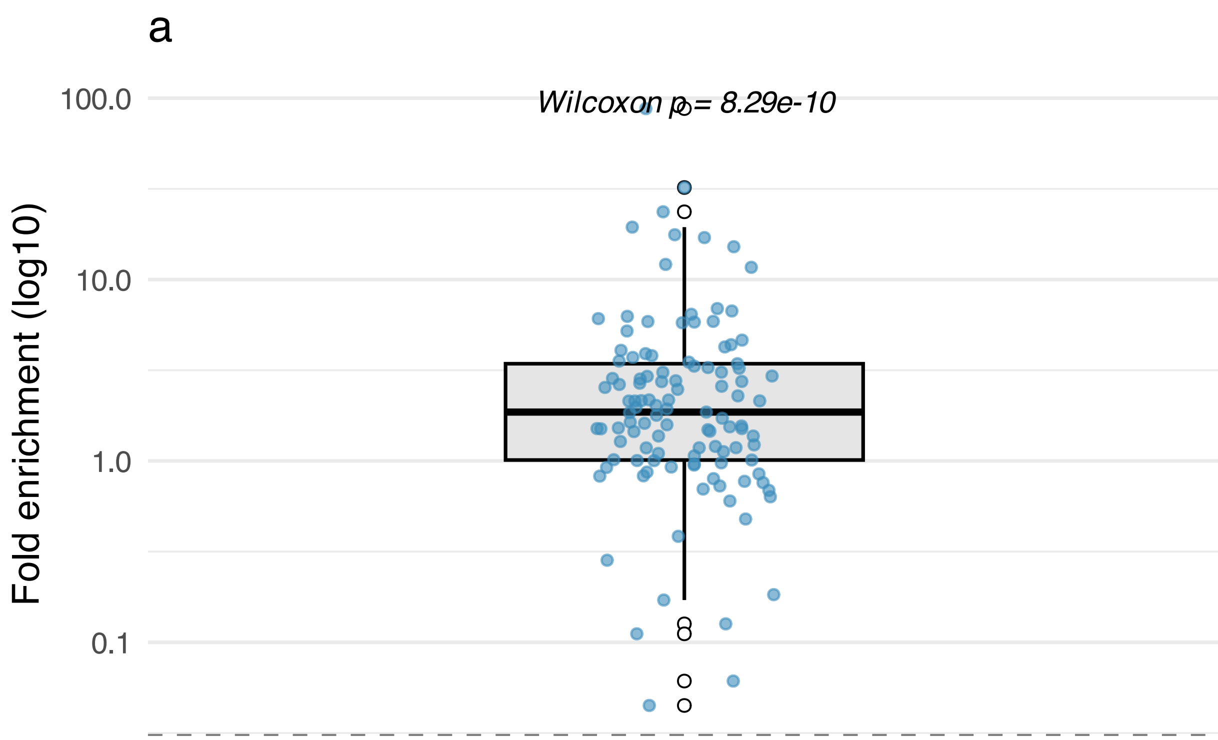
```
##  25%
## 1.01
```

```
round(quantile(overall_tbl_summary$rpm_enrichment, 0.75, na.rm = TRUE), 2)
```

```
##  75%
## 3.44
```

```r
# Paired Wilcoxon test across all samples
wilcox_res <- wilcox.test(
  overall_tbl_summary$NGS,
  overall_tbl_summary$MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)

pval <- wilcox_res$p.value

log2fc_plot = ggplot(overall_tbl_summary, aes(x = "", y = rpm_enrichment)) +
  geom_boxplot(width = 0.4, fill = "gray90", color = "black", outlier.shape = 21, outlier.fill = "white
  scale_y_log10() +
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, color = "#3C8DBC") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  annotate("text", x = 1, y = max(overall_tbl_summary$rpm_enrichment, na.rm = TRUE) * 1.1,
           label = paste0("Wilcoxon p = ", signif(pval, 3)),
           fontface = "italic", size = 4) +
  labs(
    x = "",
    y = "Fold enrichment (log10)",
    title = "a"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "none",
  )

log2fc_plot
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```




```r
# get overall genome coverage enrichment - use wilcox_summary table which includes NA as a non detectio
overall_cov_tbl = wilcox_summary %>%
  group_by(virus) %>%
  summarize(cov_change_mean = mean(percent_enrichment), .groups = "drop") # change to depth if you want

mean(overall_cov_tbl$cov_change_mean)
```

```
## [1] 9.89
```

```r
round(quantile(overall_cov_tbl$cov_change_mean, 0.25, na.rm = TRUE), 2)
```

```
##  25%
## 4.54
```

```r
round(quantile(overall_cov_tbl$cov_change_mean, 0.75, na.rm = TRUE), 2)
```

```
##   75%
## 11.84
```

```r
# Is the different in coverage values significant?
wilcox.test(
  paired_tbl$breadth_1x_NGS,
  paired_tbl$breadth_1x_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)
```

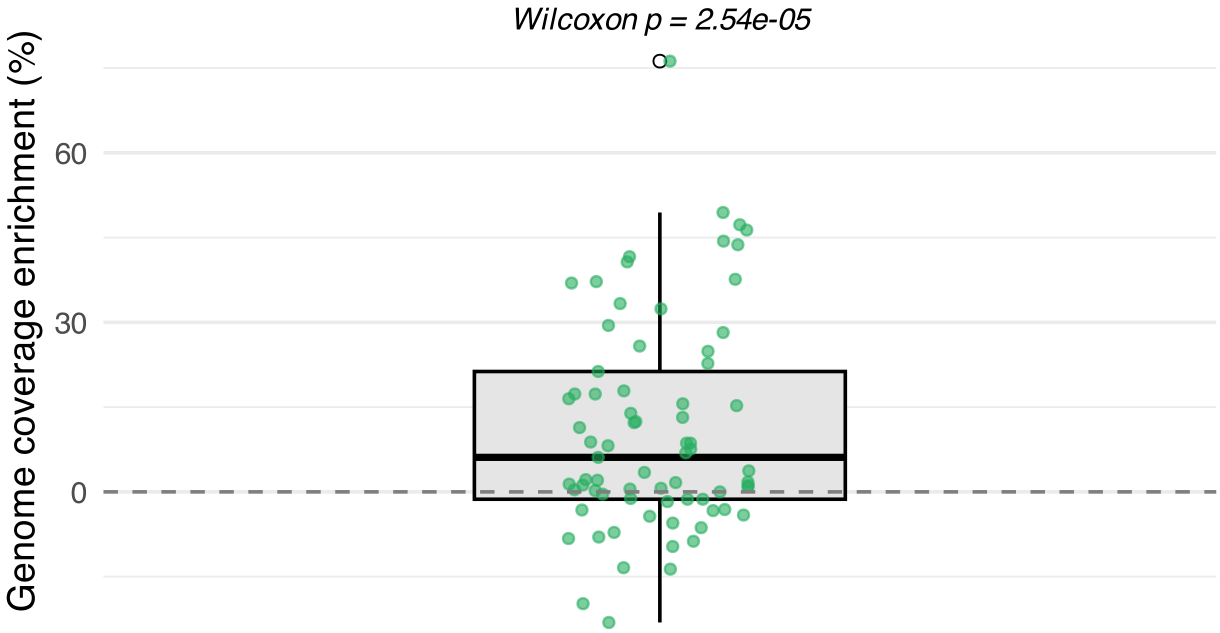
```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  paired_tbl$breadth_1x_NGS and paired_tbl$breadth_1x_MSSPE
## V = 563, p-value = 2.536e-05
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Paired Wilcoxon test across all samples
wilcox_res_cov <- wilcox.test(
  paired_tbl$breadth_1x_NGS,
  paired_tbl$breadth_1x_MSSPE,
  paired = TRUE,
  alternative = "two.sided"
)

pval_cov <- wilcox_res_cov$p.value

# summarize for boxplot
overall_cov_tbl_2 = cov_summary %>%
  group_by(virus, prep) %>%
  summarize(cov_change_mean = mean(breadth_1x, na.rm = TRUE), .groups = "drop")

# Boxplot of mean_nt_rpm by library_prep
cov_plot = ggplot(paired_tbl, aes(x = "", y = enrichment_breadth)) +
  geom_boxplot(width = 0.4, fill = "gray90", color = "black", outlier.shape = 21, outlier.fill = "white
  geom_jitter(width = 0.1, alpha = 0.6, size = 1.5, color = "#27AE60") +
  #scale_color_viridis(option = "viridis", direction = -1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  annotate("text", x = 1, y = max(paired_tbl$enrichment_breadth, na.rm = TRUE) * 1.1,
           label = paste0("Wilcoxon p = ", signif(pval_cov, 3)),
           fontface = "italic", size = 4) +
  labs(
    x = "",
    y = "Genome coverage enrichment (%)",
    title = "b"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.x = element_blank(),
    legend.position = "none",
  )

cov_plot
```

```
## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

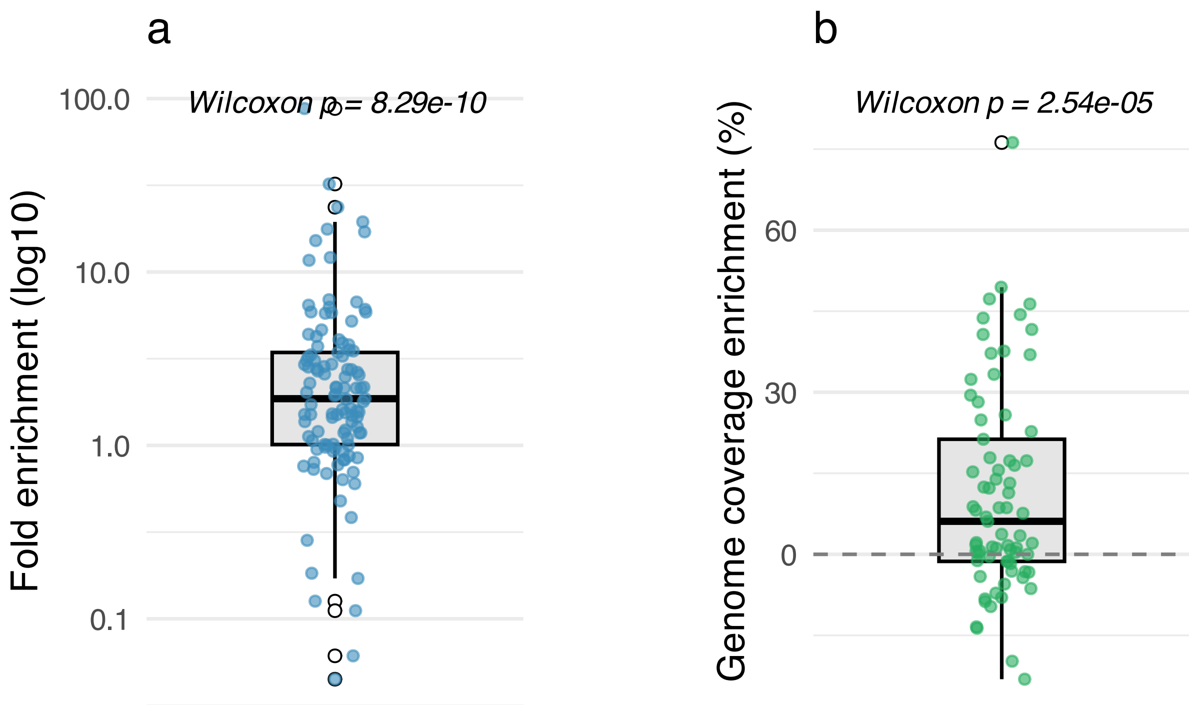
```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

b




```r
horizontal_space <- plot_spacer() + plot_layout(widths = c(0.05))

combined_boxplots = (log2fc_plot | horizontal_space | cov_plot) +
  plot_layout(ncol = 3, widths = c(1.5, 0.5, 1.5))


combined_boxplots
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.

## Warning: Removed 23 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 23 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Comparing Likelihood of detecting targets by Lib Prep or Sample Type

```
library(purrr)

# filter to targets
filter2targets = all_viruses %>%
  filter(str_detect(name_group, regex(paste0("\\b(", paste(name_targets, collapse="|"), ")\\b"), ignore_

# get tables to create matrix
all_targets = unique(filter2targets$name_group)
all_samples = unique(filter2targets$sample_name)
all_preps = unique(filter2targets$library_prep)
all_types = unique(filter2targets$sample_type)

# create library prep matrix + detected column
expanded <- expand_grid(
  sample_name = all_samples,
  library_prep = all_preps,
  name_group = all_targets
) %>%
  left_join(filter2targets %>% mutate(detected = 1),
            by = c("sample_name", "library_prep", "name_group")) %>%
  mutate(detected = ifelse(is.na(detected), 0, detected))

# do mcnemars test
results_libprep <- expanded %>%
  group_by(name_group, sample_name) %>%
  summarise(
    NGS = detected[library_prep == "NGS"],
    MSSPE = detected[library_prep == "MSSPE"],
    .groups = "drop"
```

```r
  ) %>%
  group_by(name_group) %>%
  summarise(
    mcnemar = list(mcnemar.test(table(NGS, MSSPE))),
    p_value = mcnemar[[1]]$p.value,
    # Compute discordant counts
    b = sum(NGS == 0 & MSSPE == 1),  # detected by MSSPE only
    c = sum(NGS == 1 & MSSPE == 0),  # detected by NGS only

    # Odds ratio (b/c)
    odds_ratio = ifelse(c == 0, NA, b / c),
    log_or = log(odds_ratio),
    se_log_or = sqrt(1 / b + 1 / c),
    ci_lower = exp(log_or - 1.96 * se_log_or),
    ci_upper = exp(log_or + 1.96 * se_log_or),
    .groups = "drop"
  )
```

```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```r
results_libprep
```

```
## # A tibble: 9 x 10
##   name_group mcnemar p_value     b     c odds_ratio log_or se_log_or ci_lower
##   <chr>      <list>    <dbl> <int> <int>      <dbl>  <dbl>     <dbl>    <dbl>
## 1 EV-G       <htest>  0.310     21    14       1.5  0.405      0.345    0.763
## 2 HPV18      <htest>  1         10     9       1.11 0.105      0.459    0.451
## 3 PAstV      <htest>  0.132    108    86       1.26 0.228      0.145    0.946
## 4 PBoV       <htest>  0.0286    62    39       1.59 0.464      0.204    1.07
## 5 PCV        <htest>  0.281     19    12       1.58 0.460      0.369    0.769
## 6 PKV        <htest>  0.181     10     4       2.5  0.916      0.592    0.784
## 7 PSapeV     <htest>  1         17    16       1.06 0.0606     0.348    0.537
## 8 PSapoV     <htest>  0.560     26    21       1.24 0.214      0.293    0.697
## 9 PTeV       <htest>  0.389     37    29       1.28 0.244      0.248    0.785
## # i 1 more variable: ci_upper <dbl>
```

Overall Odds Ratio

```r
# Sum discordant counts across all viruses
overall_b <- sum(results_libprep$b, na.rm = TRUE)  # MSSPE only
overall_c <- sum(results_libprep$c, na.rm = TRUE)  # NGS only

# Construct 2x2 table for McNemar
overall_tbl <- matrix(
  c(NA, overall_b, overall_c, NA),  # we only need discordant counts for OR/p
  nrow = 2,
```

```r
  byrow = TRUE,
  dimnames = list(NGS = c("0","1"), MSSPE = c("0","1"))
)

# Run McNemar test
# Note: mcnemar.test expects all 4 cells; can set concordant counts to 0
overall_tbl[1,1] <- 0  # neither detected
overall_tbl[2,2] <- 0  # both detected

overall_test <- mcnemar.test(overall_tbl)

# Overall odds ratio
overall_or <- ifelse(overall_c > 0, overall_b / overall_c, NA)
log_or <- log(overall_or)
se_log_or <- sqrt(1/overall_b + 1/overall_c)
ci_lower <- exp(log_or - 1.96 * se_log_or)
ci_upper <- exp(log_or + 1.96 * se_log_or)

overall_results <- tibble(
  odds_ratio = overall_or,
  log_or = log_or,
  se_log_or = se_log_or,
  ci_lower = ci_lower,
  ci_upper = ci_upper,
  p_value = overall_test$p.value
)

overall_results
```

```
## # A tibble: 1 x 6
##   odds_ratio log_or se_log_or ci_lower ci_upper  p_value
##        <dbl>  <dbl>     <dbl>    <dbl>    <dbl>    <dbl>
## 1       1.35  0.298    0.0870     1.14     1.60 0.000675
```

Genus Odds Ratios